

Reading *redshifts*

Use the simulation to measure shifts, calculate redshifts, and find out where the simple Doppler formula stops working — and where it gets replaced by something deeper.

Time: ~ 30 min Tool: Doppler & Cosmological Redshift simulation

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1 Measuring a *Doppler shift*

~ 8 MIN

Open the simulation. Make sure the **Classical Doppler** tab is selected. The source is moving through space and emitting wavefronts. Your job is to measure the shift, calculate z , and check yourself.

1

MEASURE

Set the source velocity to $+0.05c$ and pause the animation. The source is moving slowly toward you, so you should see a small *blueshift*... or is it a redshift? Find out for yourself.

- 1 Drag the **Source velocity** slider to $+0.05c$.
- 2 Press **Pause** when the picture is stable.
- 3 From the **Observed frame** readings under the spectrum strip, read off the wavelength of the **H α** line.
- 4 Calculate $z = \frac{\lambda_{\text{obs}} - 656}{656}$.
- 5 Open the **Measure & calculate** panel. Pick H α as the reference. Type your λ_{obs} and your z . Press **Check**.

Record λ_{obs} , z , and whether this is a redshift or blueshift. Was the sign of z what you expected?

2

MEASURE

Now reverse the motion: set source velocity to $-0.10c$. The source is now moving away from you. This time use the **H β** line (rest wavelength 486 nm).

- 1 Drag **Source velocity** to $-0.10c$. Pause.
- 2 From the observed-frame readings, read the **H β** wavelength.
- 3 Calculate $z = \frac{\lambda_{\text{obs}} - 486}{486}$.
- 4 In **Measure & calculate**, pick H β , enter your values, press **Check**.

Record λ_{obs} and z . **What's different about z this time?** What does the sign of z tell you about the source's motion?

PULL IT TOGETHER

*Write the rule for the sign of z in your own words. **What does positive z mean physically? What does negative z mean?** Refer to motion, wavelength, and frequency.*

2 When the simple formula *breaks*

~ 6 MIN

The formula on your data sheet —

$$z \approx \frac{\Delta\lambda}{\lambda} \approx \frac{v}{c}$$

— is an *approximation*. It only works when $v \ll c$. Let's find out what happens when we push it.

3

PROBE

Push the source velocity all the way to +0.9c. Read the value of z from the simulation's **Live readouts** panel.

- 1 Drag **Source velocity** to +0.90c. Pause.
- 2 Read **Redshift z** from the live readouts.
- 3 Record this value — call it z_{sim} .

Now calculate the *real* redshift using the relativistic Doppler formula:

$$z_{\text{rel}} = \sqrt{\frac{1 + \beta}{1 - \beta}} - 1, \quad \text{where } \beta = \frac{v}{c}$$

Compare z_{sim} with z_{rel} . **By what factor does the sim under-report?** What does this tell you about the linear formula at high velocities?

✓ Wave me over here — I'd like to see your z_{rel} calculation before you move on.

3 Stretched by *space itself*

~ 12 MIN

Switch to the Cosmological Redshift tab. The set-up is different now: galaxies sit at fixed positions in an expanding universe. As space stretches, photons travelling through it stretch with it. There's no Doppler-effect "source moving through space" any more — the redshift comes from the stretching itself.

4

MEASURE

Pick galaxy **G4**, launch a photon, and measure the redshift when it arrives.

- 1 Make sure $H_0 = 70$ km/s/Mpc (default).
- 2 From the **Emitting galaxy** dropdown, select **G4**.
- 3 Press **Launch photon**. Watch its colour shift as it travels.
- 4 When the photon arrives, read the observed-frame **H β** wavelength from the readings strip.
- 5 Calculate $z = \frac{\lambda_{\text{obs}} - 486}{486}$.
- 6 In **Measure & calculate**, pick **H β** , enter your values, press **Check**.

Record your λ_{obs} and z .

5

REASON

Now pick the most distant galaxy, **G1**, launch a photon, and **pause it about halfway through its journey**. Look at the photon's colour: it has already shifted, even though it hasn't reached you.

- 1 Press **Reset**. Pick **G1** from the dropdown.
- 2 Press **Launch photon**, then press **Pause** when it's roughly halfway across.
- 3 Note the photon's colour and the observed-frame line wavelengths now.

Why has the wavelength already started shifting before the photon has arrived? How is this *physically different* from the Doppler shift you measured in Phase 1? Use the words "space" and "stretched" in your answer.

6

INVESTIGATE

Hubble's law says $v = H_0 d$. **Verify this from the simulation** by checking that H_0 comes out the same for two different galaxies.

- 1 Make sure $H_0 = 70$. Pick galaxy **G2** from the dropdown.
- 2 Read **Distance** d and **Recession** v from the live readouts. Calculate $H_0 = \frac{v}{d}$.
- 3 Now pick **G5**. Read its d and v . Calculate H_0 again.

GALAXY	D (MPC)	V (KM/S)	$H_0 = V/D$ (KM/S/MPC)
G2			
G5			

Are the two values of H_0 the same? Why does this confirm Hubble's law as a universal proportionality, not just a coincidence for one galaxy?

PULL IT TOGETHER

Compare your answers to **Q1** (a moving source in Phase 1) and **Q4** (a distant galaxy in Phase 3). Both produced a value of z — but the physical reason behind each is different. Write down the difference.

4 The age of *the universe*

~ 6 MIN

The Hubble constant doesn't just give us distances — its reciprocal $1/H_0$ gives an estimate of the universe's age. Different measurement techniques give slightly different values for H_0 , and this disagreement is one of the biggest open puzzles in cosmology right now: the **Hubble tension**.

7

COMPARE

Two leading measurements of H_0 disagree:

- **Planck (CMB)** — $H_0 \approx 67$ km/s/Mpc (from the cosmic microwave background)
- **Cepheid distance ladder** — $H_0 \approx 73$ km/s/Mpc (from nearby standard candles)

Use the simulation to find the Hubble age implied by each.

- 1 Drag the H_0 slider to **67**. Read the Hubble age $\approx 1/H_0$ from the live readouts.
- 2 Drag the H_0 slider to **73**. Read the new Hubble age.
- 3 Record both values.

METHOD	H_0 (KM/S/MPC)	HUBBLE AGE (GYR)
Planck (CMB)	67	
Cepheid ladder	73	

Calculate the percentage difference between the two ages. **Why does a higher H_0 correspond to a younger universe?** Hint: think about what $1/H_0$ represents physically.

Before we regroup

If you finish early: try setting H_0 to its extremes (50 and 100) and see how dramatically the Hubble age changes. Or, in Mode A, turn on Show wavefront emission points with a moderate velocity and see if you can describe the geometric origin of the pile-up.

If you got stuck on the relativistic formula in Phase 2 or the Hubble-tension calculation in Phase 4, that's fine — bring your working to the plenary and we'll go through it together.